

Forecasting Ceres's Growth

Module 5, Part 2 (December 4)

QBA 5330

Ceres's Real Growth

- Revenue: \$24.65M (2002) → \$42.60M (2006E)
- Year-over-year growth: 8.7%, 9.3%, **19.8%, 21.4%**
- Acceleration coincides with the 2005 launch of **GetCeres™**, a dealer inventory financing program

Computing the CAGR

```
1 rev <- c(24652, 26797, 29289, 35088, 42597)
2 names(rev) <- 2002:2006
3
4 cagr <- (rev["2006"] / rev["2002"])^(1/4) - 1
5 cagr
```

```
2006
0.1465202
```

About 14.7% per year, compounded.

Base, Optimistic, Pessimistic

```
1 base_2006 <- rev["2006"]
2 base <- as.numeric(cagr)
3 optimistic <- base + 0.05
4 pessimistic <- base - 0.10
5
6 proj_2007 <- base_2006 * (1 + c(pessimistic, base, optimistic))
7 round(proj_2007, 0)
```

```
[1] 44579 48838 50968
```

A \$6.4 million spread on 2007 revenue alone.

All Scenarios Clear Breakeven

- Case breakeven: ~\$30 million in revenue
- Even the pessimistic scenario: ~\$44.6 million
- Revenue growth is not going to be the problem in this case.

The Problem: Ceres's Reporting Is Annual Only

- 5–6 data points
- No way to detect monthly seasonality or fit a moving average
- The case itself mentions demand concentrated in just 4–5 weekends each spring/summer.

Activity: A Supplementary Monthly Series

```
1 library(tidyverse)
2 library(fpp3)
3
4 set.seed(2025)
5 months <- seq(as.Date("2004-01-01"), as.Date("2006-12-01"), by = "month")
6 month_num <- as.integer(format(months, "%m"))
7 seasonal_shape <- c(0.6,0.7,1.3,1.6,1.5,1.1,0.9,0.8,0.7,0.6,0.5,0.6)
8 sales <- (700 + 12*(1:length(months))) * seasonal_shape[month_num] + rnorm(length(
9
10 sell_through <- tibble(month = yearmonth(months), sales = round(sales,1)) |> as_ts
```

Fit Holt-Winters. Forecast 6 months. When does the seasonal peak hit?

Before the Break

- Revenue is growing. Forecasts confirm it'll likely keep growing.
- But: accounts receivable **\$10.3M** → **\$14.5M**; dealer inventory **\$10M** → **\$23M** (2005 to 2006E)
- Does growing revenue automatically mean Ceres's cash position is fine?